National Institute of Technology Patna

Department of Electronics and Communication Engineering

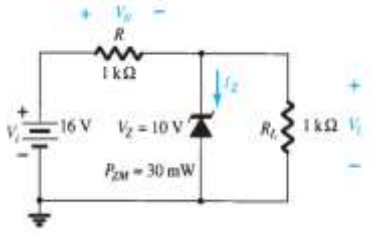
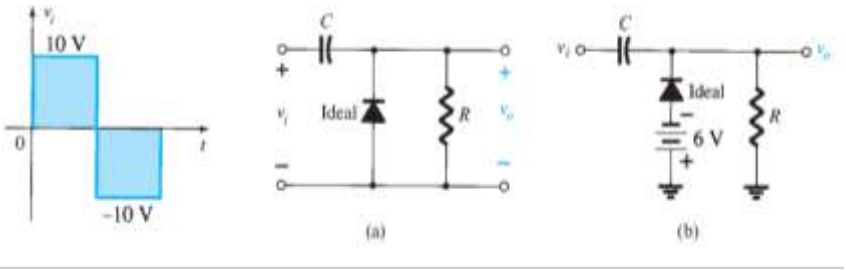
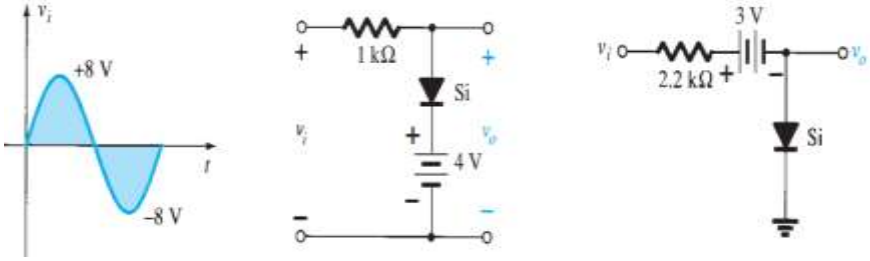
Mid Semester Exam -Subject: Elements of Electronics Engineering

(EC011301/EC011101/EC011201/EC011501/EC011401/EC15102)

Semester: 1st **MM: 30**

Course: B.Tech (EC011301/EC011101/EC011201/EC011501/EC011401/EC15102) **Time: 2.00 Hrs**

Assume missing data, if any and mention it in the answer copy

Q1.	(i) In a reverse biased p-n junction, the current is mainly due to a) Majority Carrier, b) Minority Carriers, c) Holes only, d) Electron only (ii) The potential barrier is created due to (a) Recombination of electron and hole, b) Movement of majority carriers only c) External supply voltage d) Capacitance at p-n junction (iii) Fermi level represents (a) Highest energy level at absolute zero b)The Conduction band minimum level c)The Valance band maximum level d) The forbidden energy gap (iv) At absolute zero intrinsic semiconductor behave as (a) Conductor b) Insulator c)Semiconductor d) Superconductor	4
Q2.	Explain in brief about the transition & diffusion capacitance.	3
Q3.	Draw and explain the mechanism of Full Wave Voltage Doubler.	3
Q4.	Explain working of the PN junction diode and Draw the V-I characteristics of the PN diode	3
Q5.	Explain the breakdown mechanism in diode a) Zener breakdown & b) Avalanche breakdown	3
Q6.	(A) For the Zener diode network of Fig. 1 , determine V_L, V_R , I_Z and P_Z (B) Calculate again V_L , V_R , I_Z and P_Z for same Fig. 1, while considering the $R_L = 5k\Omega$. Fig.1 	4
Q7	Sketch V_o for each network of Fig. 2 for the input shown. Fig 2. 	5
Q8	Find the Vo waveform for the both cases as shown in Fig.3 	5

*****09/10/2025(FN)****